



AI is Not Necessarily Machine Learning

The original problem definition of Artificial Intelligence was simply to deliver intelligent behavior.

How that intelligence was developed was not important; only the capability was important.

Learning, or machine learning specifically, was added by organizations that collected massive amounts of information and were looking for ways to exploit that capability for profit.

Is the focus on Machine Learning anything more than the result of the Machine Learning suppliers repeating the words “Big Data” and “Machine Learning” over and over again?

Perhaps their approach is to insert those patterns into the brains of business executives in the same way that marketing organizations repeatedly force their ideas into the public as facts.

Is it the horse driving the cart again?

Are the consulting companies promoting Machine Learning as the dominant form of AI today blindly riding that horse?

Where is the consumer of AI defining the need, and what they value, from AI?

Where is the consumer of AI demanding certain capabilities that, if they are not there, would block a particular approach?

Where is the list of AI “buying factors” that should contribute to the selection of an AI approach?

The following list itemizes some factors that should be considered (and be referenced and measured) for their value in applications requiring Artificial Intelligence (AI) behaviors. All RFQ's should be measured and evaluated for their capability importance, potential reuse, life-cycle cost/value, and schedule impact.

- identification of forms of outputs and targets of outputs from the AI
 - machines delivering safety-critical decisions and/or actions (as in autonomous weapon systems)
 - textual output to satisfy humans
- ability, not only to provide answers, but to explain why alternatives are not appropriate
- suitability for real-time adaptive control
- ability to easily create a hierarchy of expertise (systems of systems)
- capability of intelligent systems to self-organize and work as teams
- safety of systems (testable and auditable)
- ability to support handling multiple (sometimes conflicting) problems at the same time
- certification requirements (code AND application certification)
- traceable behavior (Explainable AI), including ease of use
- exposed value system for all influencing factors (a requirement for Explainable AI)
- support for after-mission reviews of production systems, including ease of use
- ability to support existing platforms/applications to extend the life of existing applications
- production memory requirements for embedded systems, if that is important
- production dependencies, if any (software / hardware)
- architecture independence (device, software application, web service)
- extensibility so the AI can easily evolve with new capabilities and new information sources
- flexibility, so one can create systems that can handle slightly different problems in slightly different ways
- adaptable, so systems can respond to situations (combinations of problems) they have never encountered before
- personalization for different users (different users may have slightly different needs)
- support for executive tuning where the person in charge may have different views of the importance of different pieces of information
- design-once-deploy-many (ability to take a behavioral policy and deploy it in multiple production platforms)
- support for human-outside-the-loop, human-on-the-loop, and/or human-in-the-loop (fixed or adaptive or as needed)
- ease-of-use / ease of understanding to justify any risks associated with a proposed solution
- user and platform pre-requisites for development (accessibility) to highlight special costs that might be incurred
- traceable responsibility / liability (for decisions and actions)
- open-source available to anyone, or proprietary where the government could acquire technology to obtain a competitive advantage

In addition, each item above should list an associated cost in schedule and money.

Summary:

It is strongly suggested that all requirements for Artificial Intelligence in solicitations should avoid dictating what AI approach must be used. Dictating what approach must be used prohibits the US from learning about new approaches to AI that could offer higher levels of service at greatly reduced costs. Requirements should be expressed in terms of required capabilities. All responses should list ALL dependencies (direct and indirect). All potential risks associated with a solution should be listed (including how any risks might be addressed: such as cost and schedule).

When new approaches to AI are developed, there should be someone within the government that is responsible for learning about those new approaches / methodologies / technologies so that they become aware of the potential benefits that can be provided, and/or the potential threats that new approaches might present if deployed by an adversary. Presently within the government, there appears to be resistance to approaches that the reviewers are unfamiliar with (cognitive dissonance).

References:

"Battlespace Automation" describes a fully automated battlespace hierarchy:

<https://www.compsim.com/news/News%20Release%20-%20Battlespace%20Automation.html>

What can one do with KEEL Technology:

<https://www.compsim.com/papers/What%20can%20one%20do%20with%20KEEL%20Technology.pdf>

Questions AI Must be able to answer:

<https://www.compsim.com/publicpapers/Questions%20ai%20must%20be%20able%20to%20answer.pdf>

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